

		Since angular velocity is inversely proportional to period, the former is smaller.
10	A	Work done in moving from X to Z $W_{X \rightarrow Z} = W_{X \rightarrow Y} + W_{Y \rightarrow Z} = 100 + (-120) = -20 \text{ J}$ $W_{X \rightarrow \infty} = W_{X \rightarrow Z} + W_{Z \rightarrow \infty} = -20 + 2000 = 1980 \text{ J}$ $\phi_X = \frac{W_{\infty \rightarrow X}}{2.0} = \frac{-W_{X \rightarrow \infty}}{2.0} = -990 \text{ J kg}^{-1}$
11	A	Heat supplied by kettle = heat absorbed by water + heat loss